Signal iOS 8.15 (build 1655)

Source-to-binary verification of the App Store build

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Result. The compiled code (`__TEXT,__text`) of all 23 binaries in the App Store app is **byte-identical** to a local build of tag `8.15.0.1655` with Xcode 26.5 (17F42). That covers Signal's five own binaries and all 18 bundled frameworks. Every other difference falls into a category already explained in the 8.14 report: signature size, UUID, linker GOT-slot order (with every stub and pointer resolving to the same target), build metadata in plists, and bitmaps rasterised at build time from identical sources.

1 Method

This is the same method as 8.35.1 and 8.14, which describe it in full.

1. The source is cloned at the Xcode Cloud path `/Volumes/workspace/repository`.
2. `make dependencies` fetches the dependencies.
3. An unsigned `xcodebuild archive` runs with the scheme's App Store Release configuration, using the exact Xcode build recorded in the shipped `Info.plist`.
4. `verify/compare_build.sh` and `verify/explain_residuals.sh` compare the result with the decrypted App Store IPA.

Input	Value
App Store app	`org.whispersystems.signal_8.15.decrypted.ipa` from BarbossHack/reproducible-LFS, SHA-256 `7629023a…6ad7e4`. Version 8.15, build 1655. Every code-signature page outside the FairPlay-encrypted ranges matches Apple's signature.
Source	Tag `8.15.0.1655`, commit `a619caa`, the commit recorded in the shipped `Info.plist`. Pods submodule `1a4acf8`.
Toolchain	Xcode 26.5 (17F42), iOS SDK 26.5 (23F73), matching the shipped `DTXcodeBuild` and `DTSDKBuild`.

2 Results

Check	Result
`__TEXT,__text`, all 23 binaries	Byte-identical (65.6 MB, hashed with Apple's `segedit` and `shasum`)
Functions, Signal's 5 binaries	144,193 of 144,193 match after address normalisation
Strings, symbols, linked libraries	Identical
Header bytes	`__LINKEDIT` size (signature) in all 23 binaries; `LC_UUID` in 13 binaries
GOT slot order	`Signal` (1,668 stub bytes), `SignalUI` (1,159), `LibMobileCoin` (4) and `SignalServiceKit` (19 relative pointers in `__const`). Every stub and pointer resolves to the same target.
Plists	26 differ, only in `BuildDetails` time, `BuildMachineOSBuild`, `CFBundleVersion` and `DTAppStoreToolsBuild`

Check	Result
Icons and asset catalogs	The two loose icons differ by at most 8/255 per channel. In the catalogs, 8 + 8 rasterised renditions differ; each comes from a vector source that is identical in both builds, or is a packed atlas of such images. The NSE catalog is identical.

These residual differences are the ones Telegram's `ipadiff.py` does not normalise. Run on this build, it flags exactly the seven files BarbossHack's 8.15 run reported: `Signal`, `SignalUI`, `SignalServiceKit`, `LibMobileCoin`, `Info.plist` and the two icons.

3 Limitations

As for earlier versions:

- The public source itself is not reviewed.
- **Prebuilt libraries: only libmobilecoin rebuilt.** This version's libmobilecoin (`signal/6.0.2`) is the same artifact that was rebuilt from source and matched byte for byte for 8.35.1 and 8.36 (8.35.1 report, section "Prebuilt native libraries rebuilt from source"). Its other prebuilt libraries, libsignal (`v0.94.4`), RingRTC (`v2.69.1`) and WebRTC (`7778b`), were checked against their pins only, not rebuilt. For the later versions 8.35.1 and 8.36, every prebuilt library reproduced exactly from public source with the projects' own recipes. That doesn't prove the same for these earlier builds, but it is a positive indicator that they too were built from their public source.
- The causes of the UUID and rasterisation differences are inferred, not proven. For the assets, re-running the logged `actool` step with the iOS 26.5 Simulator runtime matching Signal's SDK (23F73), as well as with 23F77, gave the same differences, so the runtime is ruled out; the GPU or the host macOS remain (`out/asset_runtime_test.txt`).

4 Reproducing

```
git clone https://github.com/signalapp/Signal-iOS.git /Volumes/workspace/repository
cd /Volumes/workspace/repository && git checkout 8.15.0.1655 && make dependencies
# xcodebuild archive with Xcode 26.5 (17F42), as in the 8.35.1 report, then:
verify/compare_build.sh org.whispersystems.signal_8.15.decrypted.ipa <built Signal.app>
versions/8.15.0.1655/out/ipa_vs_build
verify/explain_residuals.sh org.whispersystems.signal_8.15.decrypted.ipa <built
Signal.app> versions/8.15.0.1655/out/residuals
```

The decrypted IPA is attached to the `signal-8.15` release. The app is also in `versions/8.15.0.1655/signal_dec/`, and all outputs are in `versions/8.15.0.1655/out/`.

5 Changelog

Date	Change
7 November 2026	First published version, at `versions/8.15/`.
7 November 2026	**Moved** to `versions/8.15.0.1655/`; paths updated. **Added** the Simulator-runtime test to the limitations. **Added** this changelog.
8 November 2026	**Changed** the prebuilt-dependencies limitation: libmobilecoin is now verified (the same artifact was rebuilt from source for 8.35.1 and 8.36); libsignal, RingRTC and WebRTC for this version are not, though later versions reproducing exactly is a positive indicator.